

ML-Enabled User Stories Evaluation - Survey 1

Dear participant, This questionnaire is intended to collect data for a research project of the Software Engineering Laboratory of the Computer Science Department of the University of Salerno (UNISA) to investigate practitioners' opinions and expertise on the fairness and ethics of machine learning systems.

Regarding the survey

- The survey is structured to be completed within a maximum of 5 minutes.
- It explores personal working experiences, which may involve confidential information.
- Participation in the survey is entirely voluntary.
- You have the autonomy to stop answering and exit the survey at any time should you choose to do so.

About data:

- Your participation in this study will primarily contribute to scientific research. The results of this research will be accessible through the dissertation or scientific publications resulting from this study.
- The researchers gather social and demographic data in a manner that ensures anonymity and does not reveal personal identifying information.
- The researchers assure that any personal information obtained through this research will be treated as confidential and will not be shared or disclosed to third parties;
- This questionnaire utilizes the Google Docs app package, meaning that any information collection and utilization through it are governed by the Google Privacy Policy (<https://www.google.co.uk/policies/privacy/>).

Below are the contact details of those responsible for this survey, with which you can ask questions about your participation:

- Carmine Ferrara, Research Fellow - carferrara@unisa.it
- Francesco Casillo, Ph.D. - fcasillo@unisa.it
- Fabio Palomba, Assistant Professor - fpalomba@unisa.it (supervisor)
- Carmine Gravino, Associate Professor - gravino@unisa.it (supervisor)
- Andrea De Lucia, Full Professor - adelucia@unisa.it

Thanks in advance for your effort and time! By clicking on the "next" button, you provide your consent to the above conditions.

Salerno, July 20, 2023

Best regards.

* Indicates required question

1. Please insert your Prolific ID. *

2. By agreeing to participate in this research, I confirm that I have comprehended the objectives, risks, and benefits associated with my involvement in the study. *

Check all that apply.

☐ I Agree

ML-Specific User Stories Evaluation

Dear participant, the main idea of this survey study is to evaluate a well-defined set of synthetic ML-Specific User Stories to understand how closely they resemble those that an expert team (in the domain and ML development) would formalize during the requirements elicitation phase for a hypothetical machine learning system.

We will ask you to assess 5 random ML-Specific User Stories in terms of:

1. Comprehensibility: How easily understandable is the intrinsic text of the user story from a semantic-lexical point of view?
2. Actionability: How useful is the informational content of the user story for the implementation of a related ML-intensive system?
3. Realism: How similar is the synthetic user story compared to how it would be formalized in a professional environment?

For each of these 3 indicators, we will ask you to express an opinion from 1 to 5, where each value corresponds to:

- 1: not at all;
- 2: slightly;
- 3: neutral;
- 4: very;
- 5: extremely.

Definition of User Story

User Story

As a [user type], I want to [action], so that [benefit].

A user story is a simple and effective way of expressing a user's needs and expectations for a product or service. It helps to prioritize features and design elements that are essential to the user experience, ensuring that the final product meets the user's needs and is easy to use.

3. [Story 1] Considering the following User Story:

*

"As an educator, I want to use abstractive summarization to analyze and summarize research on effective teaching methods, so that I can apply the latest research to my own teaching practice."

How do you evaluate it. in terms of:

Mark only one oval per row.

	1	2	3	4	5
Comprehensibility	☐	☐	☐	☐	☐
Actionability	☐	☐	☐	☐	☐
Realism	☐	☐	☐	☐	☐

4. Would you formulate this user story differently? Please respond with 'Yes' or 'No'.

*

Mark only one oval.

☐ Yes

☐ No

5. If you answered 'Yes' to the previous question, could you please provide us with your revised version of the user story?

6. [Story 2] Considering the following User Story:

*

"As a literary critic or scholar, I want to use extreme learning machine to analyze literary works and predict literary trends and patterns, so that I can provide better analysis and criticism of literary works."

How do you evaluate it. in terms of:

Mark only one oval per row.

	1	2	3	4	5
Comprehensibility	☐	☐	☐	☐	☐
Actionability	☐	☐	☐	☐	☐
Realism	☐	☐	☐	☐	☐

7. Would you formulate this user story differently? Please respond with 'Yes' or 'No'.

*

Mark only one oval.

☐ Yes

☐ No

8. If you answered 'Yes' to the previous question, could you please provide us with your revised version of the user story?

9. [Story 3] Considering the following User Story:

*

"As an urban planner, I want to use preference learning to identify which urban design elements (e.g., green space, public art) are most preferred by residents in a specific neighborhood. "

How do you evaluate it. in terms of:

Mark only one oval per row.

	1	2	3	4	5
Comprehensibility	☐	☐	☐	☐	☐
Actionability	☐	☐	☐	☐	☐
Realism	☐	☐	☐	☐	☐

10. Would you formulate this user story differently? Please respond with 'Yes' or 'No'.

*

Mark only one oval.

☐ Yes

☐ No

11. If you answered 'Yes' to the previous question, could you please provide us with your revised version of the user story?

12. [Story 4] Considering the following User Story:

*

" As a movie studio, I want to use similarity learning algorithms to analyze audience data and identify similarities between different audience segments, to better understand audience needs and inform casting and marketing decisions. "

How do you evaluate it. in terms of:

Mark only one oval per row.

	1	2	3	4	5
Comprehensibility	☐	☐	☐	☐	☐
Actionability	☐	☐	☐	☐	☐
Realism	☐	☐	☐	☐	☐

13. Would you formulate this user story differently? Please respond with 'Yes' or 'No'.

*

Mark only one oval.

☐ Yes

☐ No

14. If you answered 'Yes' to the previous question, could you please provide us with your revised version of the user story?

15. [Story 5] Considering the following User Story:

*

" As a demographer, I want to use keyword spotting to identify specific demographic data in large datasets, so that I can better understand population trends and make predictions. "

How do you evaluate it. in terms of:

Mark only one oval per row.

	1	2	3	4	5
Comprehensibility	☐	☐	☐	☐	☐
Actionability	☐	☐	☐	☐	☐
Realism	☐	☐	☐	☐	☐

16. Would you formulate this user story differently? Please respond with 'Yes' or 'No'.

*

Mark only one oval.

- ☐ Yes
- ☐ No

17. If you answered 'Yes' to the previous question, could you please provide us with your revised version of the user story?

Closing

Thank you for your answers and for your precious time! Before concluding, we want to ask you some final (optional) questions. Then, you can hit "submit" to close the survey.

18. If you like, you can add other information about your experience in requirements engineering for Machine Learning Systems. Any information we did not consider is important to us!"

19. If you would like to stay updated regarding the results of our study, please leave your e-mail address.

Prolific Completion Code
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